FINAL TEST REPORT

Swag Labs — Manual QA Testing

Document version: 1.0

Project: Swag Labs

Testing Type: Manual Testing, SQL Testing, Regression Testing, Defect Verification

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1. Project Overview

This report summarizes the testing activities performed on the Swag Labs application as part of an independent Manual QA portfolio project.

The project focused on evaluating the application`s main functional areas, validating user interface behavior, checking responsive layouts, performing regression testing, verifying previously identified defects, and conducting independent SQL database testing.

The database is not connected to or representative of the application`s actual backend database.

2. Testing Objectives

The primary objectives of the testing activities were:

- Verify the main application functionality and user workflows.
- Validate authentication, logout, product catalog, shopping cart, and checkout behavior.
- Check navigation and product details functionality.
- Evaluate the application`s interface on mobile and tablet devices.

- Inspect cart-related network behavior using browser Developer Tools.
- Evaluate application behavior under simulated network conditions.
- Validate product data and database constraints using SQL queries.
- Perform regression testing to identify potential changes in previously tested functionality.
- Verify previously reported defects and document their current status.

3. Testing Scope

3.1 UI Functional Testing

The following areas were covered:

Testing Area	Description
Login	Valid, invalid, and incomplete credentials; blocked user validation
Logout	Logout functionality and access restrictions
Password Masking	Password field character masking
Product Sorting	Sorting products by name and price
Shopping Cart	Adding, removing, and managing products
Checkout	Successful checkout and empty-field validation
Social Badges	Appearance, hover behavior, and navigation
Navigation Menu	Opening, closing, and navigation links
Product Details Page	Product information and page navigation
Dynamic Catalogs	Lazy loading, spinner behavior, and slider functionality
Responsive Testing	Mobile and tablet layout validation
Network Testing	Cart-related network request inspection

Performance Testing	Application behavior under simulated network conditions

3.2 SQL Testing

SQL testing was performed using the independently created MySQL database *qa_swag_labs*.

The database contains:

- Main table: *products*
- Staging table: *products_import*
- Product records: 6
- DBMS: MySQL

The following testing areas were covered:

Data Validation

- Imported record count validation.
- NULL value validation.
- Invalid product price validation.
- Duplicate Product ID validation.
- Empty and whitespace-only product name validation.
- Duplicate product name validation.

Database Constraints Testing

- PRIMARY KEY constraint validation.
- NOT NULL constraint validation.
- CHECK constraint validation for product prices.

SQL Queries and Data Analysis

- Sorting using ORDER BY.
- Filtering using WHERE.
- Aggregate functions: COUNT, AVG, MIN, MAX, SUM.
- Keyword searching using LIKE.
- Unique value retrieval using DISTINCT.
- Repeated value identification using GROUP BY and HAVING.

Final Data Integrity Testing

- Total record count validation.
- Unique Product ID validation.
- Invalid product name validation.
- Invalid product price validation.

4. Test Environment

4.1 Devices and Browsers

The following devices and browsers are documented in the defect verification records:

Device	Browser
MacBook Air M1	Previously tested browsers: Chrome and Safari
iPhone 13 Pro	Safari Mobile
iPad A16	Safari Mobile

This documented defect verification activities included mobile and tablet testing, including iPad landscape orientation.

4.2 Tools and Technologies

Tool/ Technology	Purpose
MySQL	Database Testing

MySQL Workbench	SQL query execution and database validation
Visual Studio Code	SQL query editing
SQL	Data validation and integrity testing
Browser Developer Tools	Network inspection and network throttling

5. Test Execution Summary

5.1 UI Test execution

A total of 40 UI test cases were executed.

Metric	Result
Total Test Cases	40
Passed	40
Failed	0
Blocked	0
Not Run	0
Pass Rate	100%

All 40 UI test cases recorded a Pass status. Their Actual Results were documented as expected or consistent with the corresponding expected behavior.

5.2 SQL Test Execution

A total of 16 SQL test cases were executed.

Metric	Result
Total Test Cases	16
Passed	16
Failed	0
Blocked	0
No Run	0
Pass Rate	100%

All SQL test cases passed.

The database returned the expected results for data validation, database constraints, SQL queries, and final data integrity checks.

The database contained six product records, with no invalid product prices or product names identified during the documented checks.

5.3 Overall Test Execution

Testing Type	Total	Passed	Failed	Pass Rate
UI Testing	40	40	0	100%
SQL Testing	16	16	0	100%
Total	**56**	**56**	**0**	**100%**

The overall execution statistics above cover the primary UI and SQL test cases. Regression and defect verification results are presented separately below.

6. Defect Summary

A total of five previously identified defects were included in the defect verification activities.

All five defects remained reproducible after retesting.

Verification ID	Bug ID	Description	Issue Type	Verification Status
BV-001	BUG-1	Checkout can be competed with an empty Shopping Cart	Functional	Failed
BV-002	BUG-2	Product image is horizontally stretched on iPhone 13 Pro	UI / Visual — Responsive Design	Failed

BV-003	BUG-3	Excessive vertical spacing on the mobile Main Page	UI / Visual — Layout	Failed
BV-004	BUG-4	Footer misalignment and unintended blank space on iPad	UI / Visual — Responsive Design	Failed
BV-005	BUG-5	Login error message overflows its container and overlaps adjacent elements	UI / Visual — Error Message Layout	Failed

Defect Verification Results

Metric	Result
Total Defects Verified	5
Defects No Longer Reproducible	0
Defects Still Reproducible	5
Verification Pass Rate	0%

Interpretation:

All five previously reported defects remained reproducible during verification. No successful fixes were confirmed in the documented retesting activities.

The defects should therefore remain open until the corresponding issues are addressed and successfully verified.

7. Regression Test Summary

Regression testing was performed to verify that previously tested functionality continued to behave as expected.

7.1 UI Regression Testing

A total of 10 UI regression test cases were executed.

Metric	Result
Total Regression Tests	10
Previous Status: Pass	10
Current Status: Pass	10
Failed	0

All selected UI regression tests passed. No changes or deviations from the previous execution were observed.

7.2 SQL Regression Testing

A total of 8 SQL regression test cases were executed.

Metric	Result
Total Regression Tests	8
Passed	8
Failed	0
Pass Rate	100%

The SQL regression tests confirmed:

- Source CSV and target database record counts match: 6 records.
- No zero or negative product prices were found.
- Duplicate Product IDs were rejected by the PRIMARY KEY constraint.
- Invalid prices were rejected by the CHECK constraint.
- Product sorting returned all six records in descending price order.
- Aggregate calculations matched expected values.
- The repeated price of $15.99 was correctly identified for two products.
- Final data integrity checks confirmed six records, six unique Product IDs, zero invalid prices, and zero invalid product names.

7.3 Regression Testing Conclusion

All 18 regression test cases passed:

- 10 UI regression tests
- 8 SQL regression tests

No regressions were identified within the selected regression test coverage. Defect verification results are tracked separately and are not included in this regression pass rate.

8. Requirements Traceability

The Requirements Traceability Matrix (RTM) contains 16 derived test requirements.

Requirement Category	Total Requirements
UI Requirements	8
SQL Requirements	8
Total	**16**

All 16 requirements are marked as Covered in the RTM, with an Execution Status of Pass.

The requirements were independently derived from the available test cases because no official product requirements or formal specification were provided.

The RTM represents the documented requirement-to-test-case mapping and should not be interpreted as proof that every test case in the project has an individual requirement mapping.

9. Testing Limitations

The following limitations apply to this project:

1. No access to the actual backend database

The SQL testing activities were performed using an independently created MySQL database. The database was populated using publicly displayed product information from Swag Labs.

It does not represent or connect to the application`s actual backend database.

2. Independent SQL testing environment

SQL test results demonstrate database validation, constraint testing, query execution, and data integrity testing skills within the independently created database.

They do not confirm the correctness of the application`s internal data storage or backend implementation.

3. Derived requirements

No official product requirements or formal specifications were available. The requirements documented in the RTM were independently formulated based on the selected test cases.

4. Selected regression coverage

Regression testing was performed using a selected set of 10 UI and 8 SQL regression test cases. These results do not establish that every possible application workflow has been covered by regression testing.

5. Defects remained unresolved

Five previously identified defects remained reproducible during the documented verification activities.

10. Final Conclusion

The testing activities covered the main functional areas of the Swag Labs application, including authentication, logout, product sorting, shopping cart operations, checkout, navigation, product details, responsive behavior, network inspection, and performance-related checks.

A total of 40 UI test cases and 16 SQL test cases were executed, with all 56 test cases receiving a Pass Status.

Regression testing also produced successful: all 10 selected UI regression tests and all 8 SQL regression tests passed.

However, five previously identified defects remained reproducible during defect verification. These issues include checkout completion with an empty cart, responsive layout inconsistencies, and login error message overflow.

The independent SQL testing activities successfully validated the six-record product dataset, database constraints, query operations, and final data integrity checks.

Overall Assessment:

The documented UI and SQL test cases passed, and no regressions were identified within the selected regression coverage. Nevertheless, the five unresolved defects require further investigation and correction before the affected functionality can be considered fully validated.

End of Report